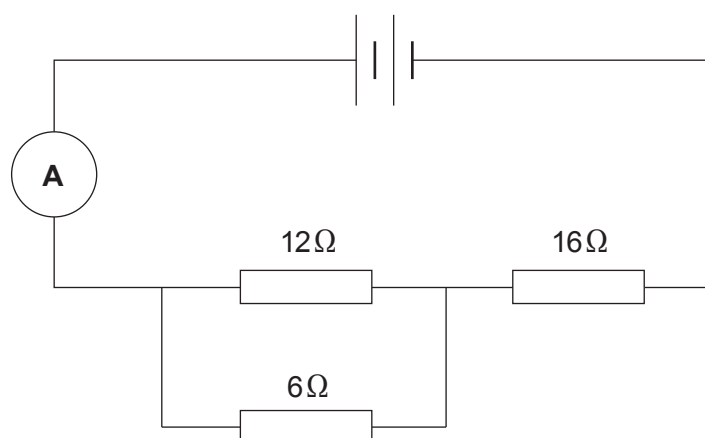


5. In an experiment a 16Ω , 12Ω and 6Ω resistor are connected as shown.



- (a) Use equations from page 2 to calculate the total resistance of the circuit.

[3]

Total resistance = Ω

- (b) A brand new battery is used in the circuit. The manufacturer states that the new battery can transfer a maximum of 75.6 kJ of energy. The current from the battery is measured as 1.5 A .

Use the equation:

$$P = I^2 R$$

and another equation from page 2, to calculate the time **in minutes** the battery will last in this circuit.

[4]

Time = minutes

